Labour Supply 3: Labour Supply Over the Life Cycle

EC306- Wages and Employment

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Goals of This Section

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Today we will:

- Show patterns of labour supply over a person's lifetime
- Discuss issues comparing labour supply of people at different ages
- Introduce lifetime labour supply model
- Discuss household production
- Analyze retirement decisions

Life Cycle Labour Supply

Introduction

Previous discussion:

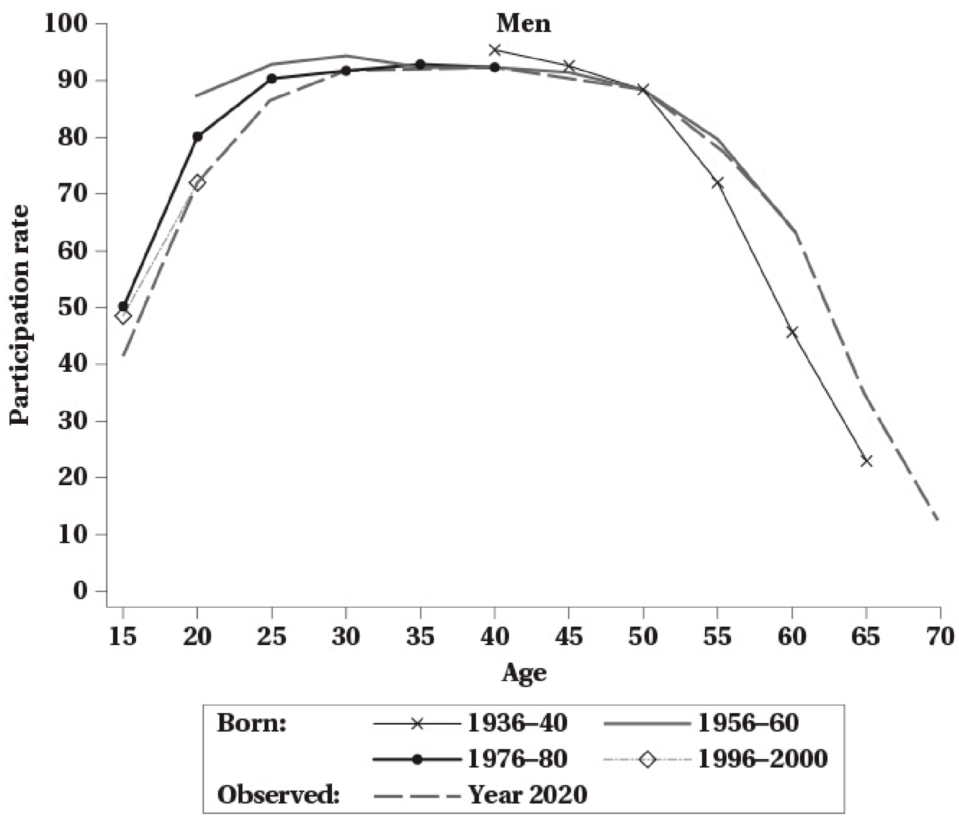
- Focused on labour supply at a moment in time
- Supply changed immediately in response to wages, non-labour income, or policies

In reality:

- People can plan work over their lifetime
- Participation is highest in prime working years
- Life events alter labour supply decisions now and in the future
  - Marriage, children, education, health, retirement

> **Tip**
>
> A life cycle approach helps us understand these patterns

Life Cycle Participation for Men



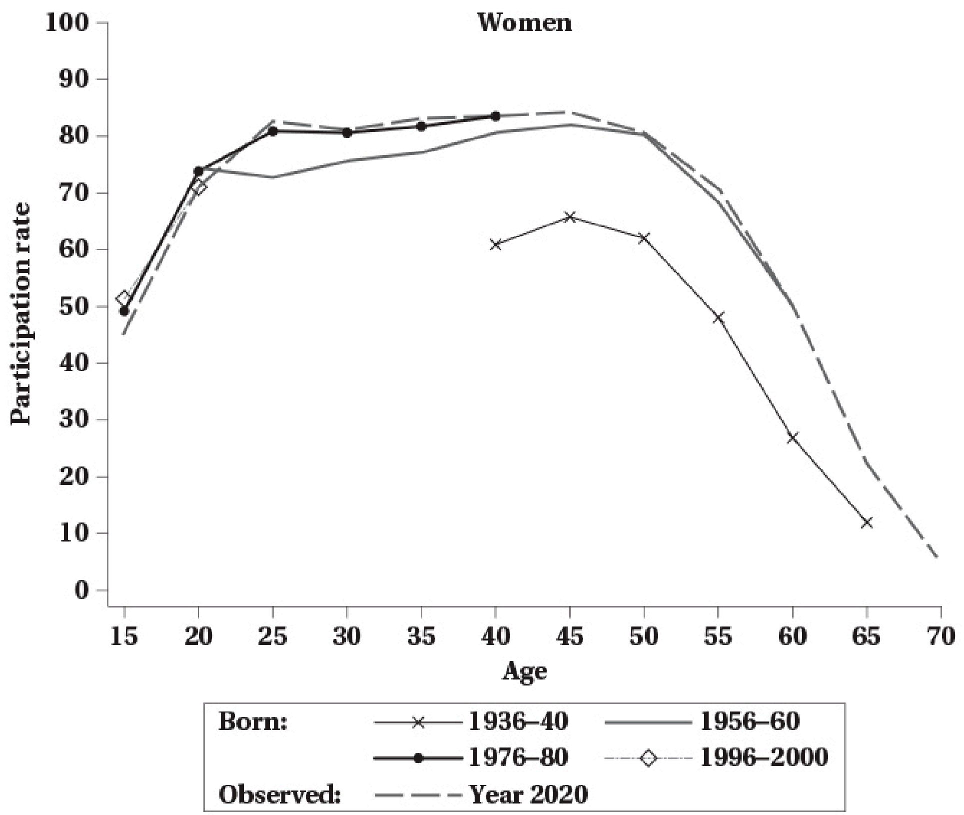
Graph follows labour supply of men born in different years

- Participation rises quickly in early 20s, peaks in late 40s, then declines

Differences across birth cohorts:

- At younger ages, younger cohorts participate a bit less
- At older ages, younger cohorts participate more

Life Cycle Participation for Women



Women follow similar lifetime pattern

- Rates are shifted down a bit relative to men
- Participation higher for younger cohorts at all ages
  - Consistent with rising female labour force participation over time

Cohort Effects vs Age Effects

The Problem with Cross-Sectional Data

When we look at participation by age in a single year, we are mixing age effects and cohort effects

Previous graphs highlight a technical point:

- Each graph contains a line that looks at life cycle pattern in year 2020
  - Participation of people in 2020 at different ages
- Problem: people at different ages are born at different times
  - 20 year olds in 2020 born in 2000
  - 60 year olds in 2020 born in 1960

Cohort Effects vs Age Effects

Definitions:

- Age effects: how labour supply changes as people get older
- Cohort effects: how labour supply differs for people born at different times

Cohort effects can be significant

- In particular for women where it has been increasing for younger cohorts
- Cannot look at participation by age in a single year and interpret as pure age effects

Better approach:

- Look at participation by age for a single birth cohort
- This is done in the graphs on the previous two slides
- Notice that the lines shift up and down for different birth cohorts
- Pattern across age is similar but there are some differences

Cohort Effects vs Age Effects

> **Practical Challenge**
>
> Very few datasets have data on the same people over their entire life

Difficulties:

- Hard to track people over time
- People die, move, refuse to respond
- Data that does track people are often restricted access

Solution: Quasi-cohorts

- Group people by birth year in cross-sectional (single year) data
- Look at how participation changes for these groups over time
- This is what the graphs on the previous two slides do
- Can be done with the Canada Census

Cohort Effects vs Age Effects

You can do this with the Canada Census:

- Census is collected every 5 years
- Each Census collects information on age and labour force status (among many other things)
- Can follow a birth cohort across different Census years as they get older

		Census	Year		
Birth Cohort	1981	1986	1991	1996	2001
1950	31	36	41	46	51
1951	30	35	40	45	50
1952	29	34	39	44	49
1953	28	33	38	43	48
1954	27	32	37	42	47
1955	26	31	36	41	46

Cohort Effects vs Age Effects

Caveat

In each Census year, people in a birth cohort are different people — not following the exact same individuals over time. This is generally accepted as a reasonable approximation.

Cohort Effects vs Age Effects

Born in 1975			
Observed in	2000	2005	2010
Age	25	30	35
Participation Rate	75%	80%	85%
Born in 1980			
Observed in	2000	2005	2010
Age	20	25	30
Participation Rate	75%	80%	85%

Illustrates problem with following ages in single Census year

- Tracks birth cohorts across Census years
- Shows age and participation rate
- For each cohort participation rises 5 points per year
- But looking at ages in a single Census year, they appear not to change

Cohort Effects vs Age Effects

Where do cohort effects come from?

People are in part defined by their generation:

- Social attitudes, cconomic conditions and policies change over time
- Generations experience different significant events (e.g. wars, disease)

We therefore expect cohorts to differ:

- Women born in early 1930s had higher labour supply than previous generations
- They were influenced by their mothers working during war
- Male children also more likely to be married to working women later in life

> **Important**
>
> Important to separate these from effects of pure aging

Dynamic Life Cycle Models

Introduction

We now know that labour supply varies over the life cycle

- The allocation of work over time can be modeled
- Models examine the effects on labour supply of wage changes

Wages can change in different ways:

- At different ages
- For a short time or permanently
- Anticipated or unanticipated

Note

Different types of changes have different effects on labour supply

Model

Individuals maximize utility over consumption and leisure over their lifetime

$$u = U(C_1, C_2, \ldots, C_N, L_1, L_2, \ldots, L_N)$$

- Here $U(.)$ is a function (e.g. logarithmic, Cobb-Douglas, etc.)

To define the budget constraint:

- We can consume and work in each period
- In theory, we can consume more than our income in a period by borrowing
- We can also save some income to consume later
- Need to be conscious of the interest rate on saving and borrowing

Model

Consider a two period model

The budget constraint says that the present value of consumption equals the present value of income:

$$C_1 + \frac{C_2}{1+r} = W_1 H_1 + \frac{W_2 H_2}{1+r}$$

- r is the interest rate for borrowing and saving
- This is the present (period 1) value formulation

Future (period 2) value formulation:

$$(1+r)C_1 + C_2 = (1+r)W_1 H_1 + W_2 H_2$$

Model

The ability to borrow and save means a person can consume more than their income in a period

Rearranging the future value of the budget:

$$C_2 = (1 + r)(W_1 H_1 - C1) + W_2 H_2$$

Two cases:

- If $W_1 H_1 > C_1$: person saved some income in period 1
  - Savings grows with interest: $(1 + r)(W_1 H_1 - C1)$ to spend in period 2
- If $W_1 H_1 < C_1$: person borrowed to consume more than income
  - $(1 + r)(W_1 H_1 - C1)$ is negative and reduces income in period 2

Model

Reexpress in terms of leisure (note that $H_t = T - L_t$):

$$C_1 + \frac{C_2}{1+r} + W_1 L_1 + \frac{W_2 L_2}{1+r} = W_1 T + \frac{W_2 T}{1+r}$$

Interpretation:

- You can consume two goods (consumption and leisure)
- Price of consumption is 1 in both periods
- Price of leisure is W_1 in period 1 and W_2 in period 2
- You consume these out of "full income"
  - Full income = income if you worked all available hours in both periods
- Period 2 values are discounted by the interest rate

Model

If you maximize utility subject to the budget constraints:

- You get a labour supply function
- Tells you optimal labour supply as a function of wages, interest rates, lifetime resources

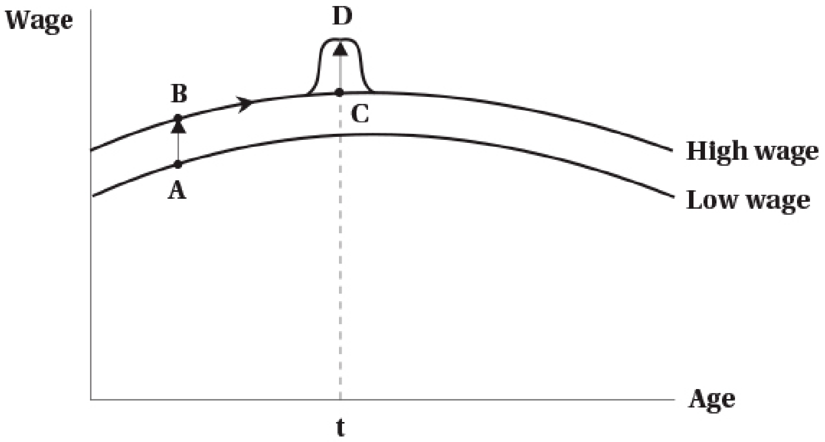
Like previous models, we can examine effects of wage changes

Three types:

1. Permanent unanticipated change in the wage in all periods
2. Anticipated future change in the wage
3. Temporary unanticipated change in the wage in one period only

These lead to substitution and income effects (more complicated because of the extra time dimension)

Permanent Wage Change



Permanent, unanticipated change in the wage:

- Increases wage in all periods
- Shifts wage schedule up

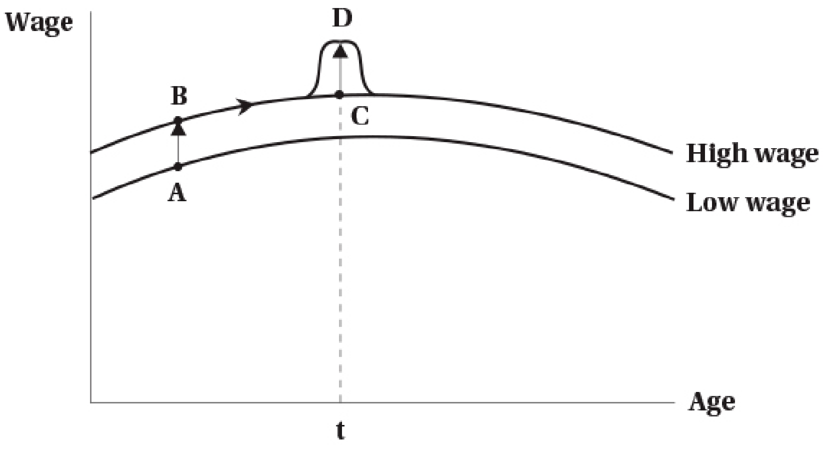
Two effects:

- Income effect: higher wage → higher lifetime income → more leisure (less work)
- Substitution effect: higher wage → leisure more expensive → less leisure (more work)

⚠ **Warning**

Labour supply effect is ambiguous

Anticipated Wage Change



An anticipated wage change moves person along their wage schedule

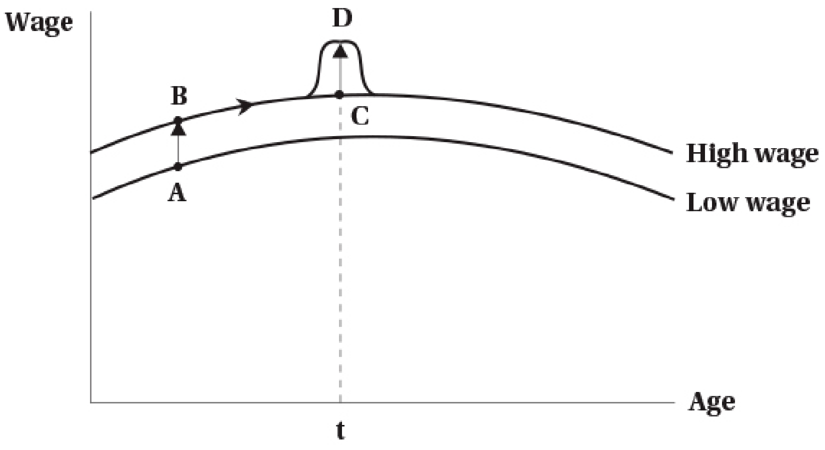
No income effect:

- Person optimized over their lifetime income
- Lifetime income has not changed

Substitution effect:

- Shifts leisure to where wage is lower
- If wage is higher in period 2, leisure is more expensive in period 2
- Less leisure in period 2 and more in period 1
- Person works more in period 2 and less in period 1

Temporary Unanticipated Wage Change



A temporary unanticipated wage change:

- Alters wage for one period (or small number) only
- e.g. bonus pay, overtime pay
- Creates a temporary blip upward in wage schedule

Two effects:

- Small income effect: slightly higher lifetime income → more leisure (less work)
- Substitution effect: leisure more expensive in period with higher wage → less leisure (more work)

Comparing Effects of Wage Changes

Strongest Effect: Anticipated Wage Change

- Pure substitution effect
- No offsetting income effect

Next Strongest: Temporary Unanticipated Change

- Substitution effect
- Small income effect
- Person works more in period with higher wage

Ambiguous: Permanent Unanticipated Change

- Substitution effect
- Larger income effect
- Ambiguous effect on labour supply

Evidence on Life Cycle Labour Supply

Interesting experiment on bike messengers in Zurich:

- Messengers paid a percentage of daily revenues
- Can choose their own total work hours

Research design:

- Researchers randomly assigned messengers to:
  - Group A: 25% higher wage for one month in September
  - Group B: 25% higher wage for one month in November
- Each acted as control group for the other
- Both get same increase in lifetime income, so difference is substitution effect

Finding

Messengers shift labour supply to periods with higher wage

Household Production

Introduction

Previous Assumption

People choose between consumption and leisure, where leisure = time not working for pay

In reality, people also spend time on household production:

- Cooking, cleaning, childcare, home maintenance, shopping, etc.
- Produces goods and services that can be consumed
  - Meals, clean clothes, child supervision, etc.

In this section we explicitly model household production

Model

People gain utility from the consumption of goods and services:

$$u = f(Z_1, Z_2, \ldots, Z_N)$$

- $Z_1, Z_2, \ldots, Z_n$ are final goods and services
- $f(.)$ is a function (e.g. Cobb-Douglas, logarithmic, etc.)

Each good/service is produced by the household using:

- A set of market inputs $X_1, X_2, \ldots, X_n$
- Time h_i

Time is divided between:

- Market work
- Household production tasks

Model

What does it mean that households “produce” the goods they consume?

Does not mean that literally everyone is producing goods from raw materials

Instead:

Every good/service consumed requires some time to prepare

- Hiring contractors to remodel a home requires a small amount of home production
  - Time spent explaining what you need, keeping contractor on schedule
- Remodelling your basement by yourself requires much more time

In this model:

We explicitly account for substituting time between various uses:

- Work, home production of various goods, leisure

Model

> **Example: Household Consumes Two Goods**
>
> Nutrition and rest — These would be Z_1 and Z_2 in the utility function

Nutrition is produced using:

- Processed or raw food, and time
- These are X_1, X_2 and h
- Raw food takes time to prepare equal to $h = d$ hours

Each of the three purchased goods has a price:

Processed food:

- Price per unit of nutrition is P_c
- Takes no time to prepare

Model

Raw food:

- Price per unit of nutrition is $P_r + W_d$
- The added time cost is valued at the wage rate W times d hours

Leisure:

- Price is the wage rate W

Comparative Statics

What happens when we start changing some of the prices?

Increase in wage rate W:

- Cost of time affects both rest and production of raw food
- Higher wage makes home production more expensive
  - Time spent cooking becomes more expensive
  - Substitute away from home production to processed food
  - Substitution effect: less home production, more processed food
  - Income effect: higher wage → higher income → more consumption of all goods
- Effect on rest is same as before
  - Substitution effect: less rest, more work
  - Income effect: higher income → more rest

Comparative Statics

Decrease in price of processed food P_c:

- Processed food becomes cheaper relative to other goods
- Substitution effect: more processed food, less home production
- Income effect: lower price → higher real income → more consumption of all goods

Decrease in price of raw food P_r:

- Raw food becomes relatively cheaper
- Substitution effect: more raw food, less processed food
- Income effect: lower price → higher real income → more consumption of all goods

Comparative Statics

Decrease in time required to prepare raw food d:

- Example: technology that reduces cooking time (e.g. Instant Pot)
- Home production becomes relatively cheaper (through lower time cost)

Effects:

- Substitution effect: more raw food, less processed food
- Additional substitution effect: higher real wage (keep more of their wage) → less rest, more work
- Income effect: lower time cost → higher real income → more consumption of all goods

Discussion

Model adds more real world complexity:

- Matches what people actually do
- Emphasizes that people do not just choose between work and leisure
- Also emphasizes the role of the household relative to the individual
  - Household members can specialize in different tasks
  - Changes for one household member can affect others

> **Household Production Function**
>
> Different households have different abilities to produce goods. This can affect their choices and responses to changes in prices.

Fertility and Childbearing

Introduction

Fertility is a key life event that affects labour supply

- Particularly for women, but also men to some degree
- Children are typically planned to some degree
  - Timing of children is often chosen to fit with work plans
  - Number of children is often chosen based on lifetime work plans

Economists have developed models to analyze fertility decisions

- These models are similar to the life cycle labour supply model
- We will not go over the model (beyond the scope of this course)
- But we will discuss some of the key variables that affect fertility decisions

Variables Affecting Fertility - Income

Theoretical Prediction

Income is positively related to the number of children — higher income → higher ability to pay for children. Children are a "normal good."

Some disagreement on whether children are considered a normal good:

- Empirically, higher income families and countries have lower fertility rates
- Explanation: substitution effect of higher income dominates income effect
  - Higher income → higher opportunity cost of time spent on children

Variables Affecting Fertility - Price of Children

Like goods, the demand curve for children slopes down

- Higher “price” of children → fewer children are chosen

Price of children includes:

- Direct costs: food, clothing, housework, etc.
- Opportunity costs: time spent caring for children could be spent working for pay

A higher price (wage rate) creates income and substitution effects:

- Income effect: higher price → higher lifetime income → more children
- Substitution effect: higher price → time with children more expensive → fewer children

> **Note**
>
> Effect is theoretically ambiguous, but empirically substitution effect dominates

Variables Affecting Fertility - Price of Related Goods

Many goods are consumed as part of having children:

- Childcare, education, etc.
- These are complementary goods
  - If you have more children, you will consume more of these goods
  - You could also consider these direct costs of children

Economic Principle

An increase in the price of a complementary good reduces demand. Higher price of childcare or education → fewer children.

Variables Affecting Fertility - Tastes

Tastes for children vary across people and over time:

- Some people really want children, others do not
- Some cultures value children more than others
- Attitudes towards children have changed over time

A stronger taste for children means more children

- This is like a shift in the demand curve for children

Over time in Canada:

- There has been a shift away from children
- Fertility rates have declined over time
- People are having fewer children than in the past

Variables Affecting Fertility - Technology

Technology in this case could mean several things:

- Could refer to the "production" of children
- Might also relate to the goods and services used with children (e.g. home automation, disposable diapers)

Key developments in the last few decades:

- Birth control: allows people to better plan timing and number of children
- Fertility treatments: allows some people to have children who otherwise could not
- Medical care: reduces infant mortality, so fewer children needed to ensure some survive
- Home technology: reduces time cost of childcare, so lower opportunity cost of children

> **Tip**
>
> Higher technology in any of these areas tends to increase demand for children

Empirical Evidence

Economists have examined the relationship between income and fertility

> ⚠ **Challenge: Omitted Variables**
>
> A simple correlation between income and fertility tends to be negative. Important to hold constant other variables that affect both income and fertility.

Example:

- Education levels are positively related to income and negatively related to fertility

When other variables are held constant:

- Income tends to be positively related to fertility
- Consistent with children being a normal good
- Tax incentives also tend to increase fertility

Retirement and Pensions

Introduction

Retirement is a key life event that affects labour supply

- Most people retire at some point
- Retirement age has been changing over time

It can mean many things:

- Permanent exit from the labour force
- A reduction to part-time work
- Retiring from one full-time job and taking another

For most people:

Retirement means permanent exit from the labour force

Determinants of Retirement - Age

> **Most Important Determinant**
>
> Age is the most important determinant of retirement

- Most people retire between ages 60 and 70
- Very few people retire before age 50 or after age 75

Mandatory retirement:

- In Canada it is generally illegal to force retirement based on age
- Except if deemed a bona fide occupational requirement (BFOR)
  - Like for pilots, police, firefighters, air traffic controllers
- Still exists in other parts of the world (e.g. Japan, China, parts of Europe)

Determinants of Retirement - Wealth/Earnings

Wealth in our model entered as non-labour earnings:

- Higher wealth → higher non-labour income
- Higher non-labour income → more leisure (less work)

Earnings changes have income and substitution effects:

- Income effect: higher earnings → higher lifetime income → more leisure (less work)
- Substitution effect: higher earnings → leisure more expensive → less leisure (more work)

Can also think of this in a lifecycle model

Determinants of Retirement - Other

Health is an important determinant:

- Poor health can force people to retire early
- Good health can allow people to work longer
- Health of family members can also play a role
- Might operate through non-labour income or preferences

Changing nature of work:

- Move towards less physically demanding jobs can prolong work life
- More flexible work arrangements can allow people to work longer
- Alternatively, more stressful jobs can lead to earlier retirement
- Automation might force early retirement
- Could affect the budget constraint or preferences

Determinants of Retirement - Other

Changing nature of family:

- Many more two-earner families now
  - Can lead to earlier retirement for one spouse
  - Or could prolong work if spouses want to work together longer
- Children living at home for longer can affect retirement timing
- Increase in caring for elderly parents can lead to earlier or later retirement
- Might affect budget constraint or preferences

Retirement Income

> **Major Effect on Retirement Decisions**
>
> Public and private pension plans determine most income in retirement

The pension system in Canada consists of:

1. Old Age Security (OAS)

- A flat rate pension available to most Canadians over age 65
- Funded out of general tax revenues
- Monthly max is $740 for those aged 65-74, $814 for those 75+
- Clawback for high income earners (above $148,000 in 2024)
- Supplemented by the Guaranteed Income Supplement (GIS) for low income seniors

Retirement Income

2. Canada/Quebec Pension Plan (CPP/QPP)

- Earnings-related pension available to those who have contributed
- Funded by contributions from employees, employers, and self-employed
- Monthly max is $1,433 in 2025 (age 65)
- Can start receiving as early as age 60 (reduced) or as late as age 70 (increased)
- Based on average earnings over working life, up to a maximum limit

3. Employer Pension Plans

- Many employers offer pension plans to employees
- Can be defined benefit or defined contribution plans
- Vary widely in terms of benefits and eligibility
- Includes private savings from RRSPs

Old Age Security

What effect does OAS have on retirement decisions?

Operates as a lump sum transfer:

- Increases non-labour income
- Creates income effect: more leisure (less work)
- No substitution effect because it does not change the wage rate

GIS operates like a negative income tax:

- People get the lump sum payment
- Payments reduced by $0.50 for each dollar of work income
- Has work incentives equal to a negative income tax

Canada/Quebec Pension Plan

The effect of pension income on retirement depends on how it is structured

In Canada currently, it is simply a fixed payment:

- Increases non-labour income
- Creates income effect: more leisure (less work)
- No substitution effect because it does not change the wage rate

In the past, there was a provision that reduced benefits if you worked:

- Similar to a negative income tax
- If the tax back rate is 100%, then it is like pure welfare
- Could also not tax back until a certain income level

Employer Pension Plan

Generally involve a fixed payment based on years of service and earnings

- Increases non-labour income
- Creates income effect: more leisure (less work)
- No substitution effect because it does not change the wage rate

Over time the parameters that determine the fixed payment can change:

- This can alter work incentives at retirement
- Example: if the benefit formula is changed to require more years of service
  - Would reduce the income effect of the pension
  - Could also create a substitution effect to work more to get the pension

Evidence on Effect of Pensions

Main Finding

Studies find that pensions affect retirement in expected ways

- Pensions that increase non-labour income lead to earlier retirement
- Pensions that create work incentives lead to later retirement

More recent work:

- Increase in retirement age increases working-age disability claims in UK
- Increasing public pension participation improves health behaviours (less smoking)
- Pension reform allowing Japanese spouses to claim half of pension earnings increased divorce

Summary

Summary

Labour supply varies over the life cycle

- Participation highest in prime working years
- A life cycle approach helps us understand these patterns

Dynamic life cycle models can analyze effects of wage changes:

- Permanent unanticipated changes
- Anticipated future changes
- Temporary unanticipated changes

Household production is an important part of labour supply decisions

- People allocate time between work, household production, and leisure
- Changes in prices of goods and time costs affect these choices

Summary

Fertility decisions are affected by:

Income, prices, tastes, and technology

Retirement decisions are affected by:

Age, wealth, health, family, and pensions